Thirty-Five Worlds

Team Members: ________________ ________________ ________________

________________ ________________

Seven patients. Three got nothing, four got the new treatment, and we recorded how many days each took to recover.

	Control (3)	**Treated** (4)
recovery, days	22, 33, 40	19, 22, 25, 26
mean	31.67	23.00

The treated group recovered 8.67 days sooner on average. That looks like a result. But one untreated patient recovered in 22 days, as fast as anyone who was treated, and with seven people the whole thing could be an accident of who landed in which group.

Null hypothesis: the treatment did nothing. If that is true, then these seven recovery times were going to be what they were regardless, and which three people we called "control" was arbitrary. So look at every way it could have come out.

Part 1. Thirty-Five Worlds

20 minutes

There are exactly 35 ways to pick 3 patients out of 7. Your slip has some of them. Between you, the room has all of them — every world the null hypothesis allows.

1.1 Your team's sums, from the slip:

________ ________ ________ ________ ________ ________

1.2 ***Class check before anything else.*** **Every patient appears in exactly 15 of the 35 splits, so all 35 sums must total $\boxed{15 \times 187 = 2{,}805}$.**

Board total: ____________ **Match? YES NO — find the error before going on.**

Each sum S is the total recovery time of a control group. Turning it into a difference in means takes no more arithmetic than this:

$$\text{difference} = \frac{S}{3} - \frac{187 - S}{4} = \frac{7S - 561}{12}$$

1.3 The real experiment had control patients P1, P2, P3, so $S = 95$. Check that the formula returns the observed difference: ____________

Somebody's slip contains the actual experiment. Whose? ____________

Say why it *has* to appear among the 35, and what that means about where the observed result sits relative to the null distribution.

1.4 Count the two tails of the board separately.

Difference ≥ 8.67, that is $S \geq 95$ — the treated group did better than we saw: ________

Difference ≤ -8.67, that is $S \leq 63$ — the *control* group did that much better: ________

one-sided $p =$ _____ $/35 =$ _________ **two-sided** $p =$ _____ $/35 =$ _________

1.5 One-sided asks whether the treatment helped. Two-sided asks whether it did anything at all, help or harm. Which would your team have committed to *before* seeing the data, and why? (Notice that you are only allowed to answer this honestly once.)

1.6 Doubling the one-sided p-value does not give you the two-sided one. Look at the range of differences on the board — the largest and the most negative — and explain why not.

1.7 Neither p-value is an estimate. Each is the exact probability of a result this extreme if the treatment did nothing, and you got them by adding three numbers at a time. At $\alpha = 0.05$, what do you conclude — in a sentence that does not use the word "proves"?

Part 2. Why Bother Simulating?

12 minutes

Open `Class15_Thirty_Five_Worlds_Skeleton.ipynb`. You just did by hand what the notebook is about to do three ways.

> **2.1 Exhaustive enumeration, in code. Does it agree with the board?** ____________
>
> **2.2 Five thousand random shuffles instead.** $p \approx$ ____________
>
> **Run it twice more:** ____________ ____________

Seven patients gave 35 splits, which a room full of people got through in ten minutes. A realistic trial — 70 patients, 30 control and 40 treated — has

$$55{,}347{,}740{,}058{,}143{,}507{,}128 \text{ splits, or about } 5.5 \times 10^{19}$$

> **2.3 Suppose a computer could check a million of those every second. How long would the exhaustive version take?** ____________________________
>
> **An order of magnitude and a unit anyone can picture is enough. Then say, in one sentence, what that number is an argument for.**

Part 3. What a p-value Does When Nothing Is Happening

13 minutes

The notebook now runs 500 complete experiments in which the treatment genuinely does nothing — no effect, by construction — and records the p-value from each.

> **3.1 *Predict first.* Most of those 500 p-values will be:**
>
> **NEAR 1** **NEAR 0.5** **NEAR 0** **SPREAD EVENLY**

> **3.2 Describe the histogram you got in one sentence.**
>
> **Fraction of the 500 with** $p < 0.05$**:** ____________

> **3.3 Nothing was wrong with any of those 500 experiments. No bad data, no mistakes, no cheating. Explain, in terms of the histogram, where those significant results came from.**

3.4 Now the same 500 experiments with a real effect present.

Fraction with $p < 0.05$: ___________

A real effect was there every single time, and the test still missed it in a good fraction of the runs. Which type of error is that, and what would you change to make it rarer?

Discussion

5 minutes

D1. Thirty-three groundhogs predict the weather and Essex Ed comes out significant. You have now seen the machinery that produces an Ed. Using your Part 3 histogram, explain to someone who has not taken this course why finding one significant groundhog out of thirty-three is not evidence of anything.

D2. Our seven-patient trial did not reach significance. A colleague writes up the result as "the treatment was shown to have no effect." Two things are wrong with that sentence. Name both.